

CSE 208: Data Structures and Algorithms II

Online on Flow & Matching

Department of Computer Science and Engineering

Term: July 2025

Time: 45 Minutes

It is the start of the semester, and the university advising portal is crashing. The administration has introduced a flexible "Team Teaching" system.

There are P **Students**, Q **Teachers**, and R **Courses**.

- **Students:** Each student needs to complete a specific number of credits.
- **Courses:** Each course is worth 1 credit and has a strictly limited **Seat Capacity** (maximum number of students that can be enrolled in it).
- **Teachers:** A course can be taught by **one or more** teachers who share the student load. However:
 - A teacher can only teach courses they are qualified for.
 - Each teacher has a **Student Limit**: the maximum total number of students they can supervise across *all* the courses they teach combined.

Task Requirements

Determine if it is possible to assign students to courses, and distribute the course load among the available teachers, such that:

1. Every student meets their credit requirement.
2. No course exceeds its seat capacity.
3. No teacher exceeds their personal student limit.
4. Teachers are only assigned to courses they are qualified for.

Input Format

- The first line contains three integers P, Q, R (Students, Teachers, Courses).
- The second line contains P integers, where the i -th integer is the required credits for Student i ($1 \leq i \leq P$).
- The third line contains Q integers, where the j -th integer is the max student limit for Teacher j ($1 \leq j \leq Q$).
- The fourth line contains R integers, where the k -th integer is the seat capacity for Course k ($1 \leq k \leq R$).
- The next line contains an integer K (number of eligible Student-Course pairs).
- In the next K lines, there will be two integers u, v , meaning Student u is eligible to take Course v ($1 \leq u \leq P, 1 \leq v \leq R$).

- The next line contains an integer M (number of compatible Teacher-Course pairs).
- In the next M lines, there will be two integers t, c, meaning Teacher t is qualified to teach Course c ($1 \leq t \leq Q$, $1 \leq c \leq R$).

Output Format

- If a valid assignment exists for all students, print **YES** followed by the total number of course assignments made.
- Otherwise, print **NO**.

Constraint

Runtime must be $O((P+Q+R) \cdot (K+M)^2)$.

Sample I/O

Input	Output
2 2 1 1 1 1 1 2 2 1 1 2 1 2 1 1 2 1	YES 2